

CSE 112 Lab #5 – Chapter 15 (page 985 - #12) – Derived Classes, and Accessors and Mutators

Use the Ship class below and derive a Cargo and Cruise ship class from the ship class IAW chapter 15 Programming Challenges # 12 on page 985 of the text.

The ship classes should be declared in a separate header file. Demonstrate the class with a program (shown below):

```
int main()
{
    // Create an array of Ship pointers, initialized with the addresses of 3 dynamically allocated objects.
    Ship *ships[3] = { new Ship("Lolipop", "1960"),
                      new CruiseShip("Disney Magic", "1998", 2400),
                      new CargoShip("Black Pearl", "1800", 50000)
                    };

    // Call each object's print function.
    for (int index=0; index < 3; index++)
    {
        ships[index] -> print();
        cout << "-----\n";
    }
    return 0;
}
```

Grading will be based on meeting all of the lab requirements, adherence to programming standards, operation and accurate output, and programming style including white space and indentation.

```
// Ship class
class Ship
{
private:
    string name;    // Ship name
    string yearBuilt; // Year it was built

public:
    // Constructor
    Ship(string n, string y)
    {
        name = n;
        yearBuilt = y;
    }

    // Accessors
    string getName()
    { return name; }

    string getYearBuilt()
    { return yearBuilt; }

    // print function
    virtual void print()
    {
        cout << "Name: " << name << endl
              << "Year built: " << yearBuilt << endl;
    }
};
```